Bhavana Sallagundla Jayappa

ML Engineer | Junior Data Scientist | AI Engineer | Willing to relocate

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EXPERIENCE

Georgia State University — Atlanta, GA
Graduate Assistant — Aug 2024 – Present

- Built and maintained weekly data pipelines using **Python, Pandas and PostgreSQL** to clean & validate 15,000+ student records, performing checks, null-value filtering, duplicate detection, and outlier analysis, reducing data errors by 20%.
- **Segmented student cohorts** using **K-means clustering,** Silhouette scores based on academic and behavioral data to identify groups with distinct learning patterns, supporting targeted academic guidance and program planning.
- Developed and automated weekly **KPI dashboards** using **Tableau and Excel**, summarizing key metrics for faculty and leadership, reducing manual reporting time by 30% and enabling timely monitoring of student performance.
- Analyzed student assessments, lab work, and exam results for **70+ students** to identify learning gaps and provide targeted guidance, helping struggling students improve understanding and engagement.
- Facilitated **instructional support** through labs and review sessions**,** grading assignments and offering one-on-one feedback, contributing to **a 10% increase in assignment completion rates**.

Tata Consultancy Services Ltd — Bengaluru, India
Machine Learning Engineer — Jun 2022 – Aug 2024

- Performed large scale **data wrangling and exploratory analysis**, transforming raw datasets into structured, high-quality inputs and uncovering complex trends that informed feature engineering and modeling strategy.
- Developed and optimized predictive models using **cross-validation, hyperparameter tuning, confusion-matrix evaluation**, logging experiments in MLflow & reducing iteration cycles by 30% through reproducible, version-control.
- Established end-to-end **ML pipelines** on **AWS (S3, SageMaker)** integrated with **MLflow**, enabling seamless **model versioning** and **deployment.**
- Collaborated with cross functional teams to integrate trained models into production via APIs, real-time inference endpoints, to monitor continuously and automated retraining triggers, reducing manual work by 20%.

Agimus Technologies — Bengaluru, India
Machine Learning Intern — June 2021 – Aug 2021

- Designed and implemented **web scraping pipelines** to collect and consolidate structured data from multiple online sources, building a clean dataset for downstream analytics and model development.
- Performed **exploratory data analysis** using **Python, Pandas, NumPy, Seaborn, Matplotlib** to identify trends, outliers, and feature relationships, guiding feature selection and modeling strategy.
- Developed and evaluated **classical ML models** including Linear Regression, Logistic Regression, and Random Forest, comparing performance metrics to select optimal models for predictive analysis.

PROJECTS

Bank Loan Default Prediction

- Conducted EDA on **100K+ borrower records**, performing data cleaning, transformation, and feature engineering to ensure high-quality inputs for ML modeling and credit-risk predictions.
- **Identified key patterns and correlations** in credit-risk forecasting, linking credit scores to borrower behavior, and generated insights through advanced data visualizations that improved risk management dashboards.
- Built and optimized **Random Forest and XGBoost models**, leveraging cross-validation, and hyperparameter tuning to achieve 96% accuracy in predicting high-risk borrowers, enhancing decision-making for lending strategies.

EDUCATION

Georgia State University, Atlanta, GA — Aug 2024 – May 2026 (Expected)
Master of Science, Data Science and Analytics — GPA: 4.0

Dr. Ambedkar Institute of Technology, Bengaluru, Karnataka — Aug 2018 – July 2022
B. Tech in Electronics & Communication Engineering — GPA: 8.7/10

SKILLS

Programming & Frameworks: Python, Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, SQL, MySQL
Machine Learning: Feature Engineering, clustering, anomaly detection, predictive modeling, classical ML, DL
MLOps & Engineering: ML pipelines, ETL & ELT, SageMaker, Data Wrangler, A/B testing, Hyperparameter Tuning
Cloud & Analytics: AWS (Lamda, S3, EC2, Glue, IAM, Kinesis). Tableau, Power BI, KPI Tracking, Data visualization

CERTIFICATIONS:

AWS Certified Machine Learning - Associate https://www.credly.com/badges/73ae7d77-174c-4687-aaec-26da20b65173
Data Engineer by DataCamp - https://www.datacamp.com/completed/statement-of-accomplishment/track/55519f12806d16637d7fbfd1770d4115cb7857db

PUBLICATIONS:

Bhavana S J, Spoorthi P A, Anitha G S, Jayashree A M "**Classification of Rail Track Defects Based on Computer Vision Using DNN",** DOI Link: https://doi.org/10.22214/ijraset.2022.45968